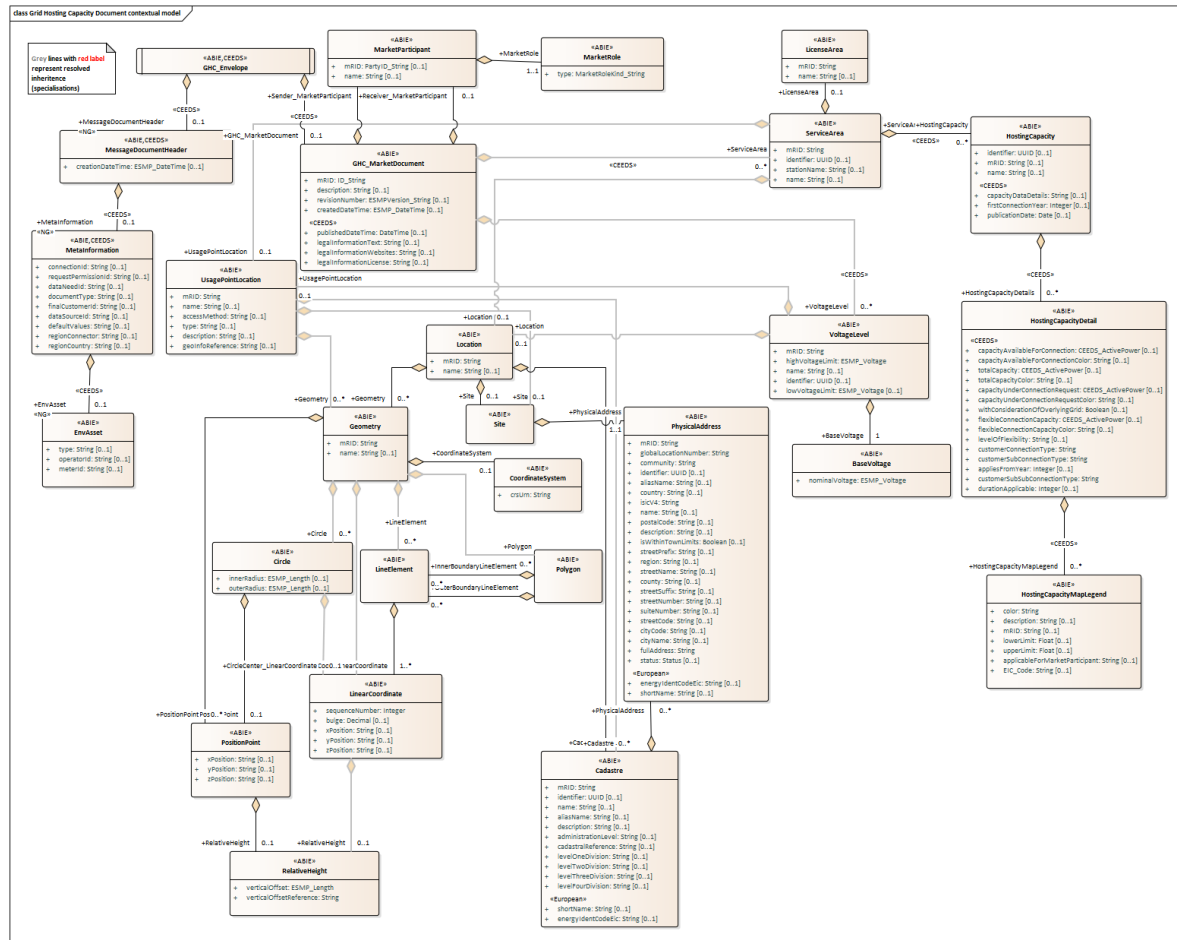


## Grid Hosting Capacity Document contextual model

## Overview of the model

!# shows the model.



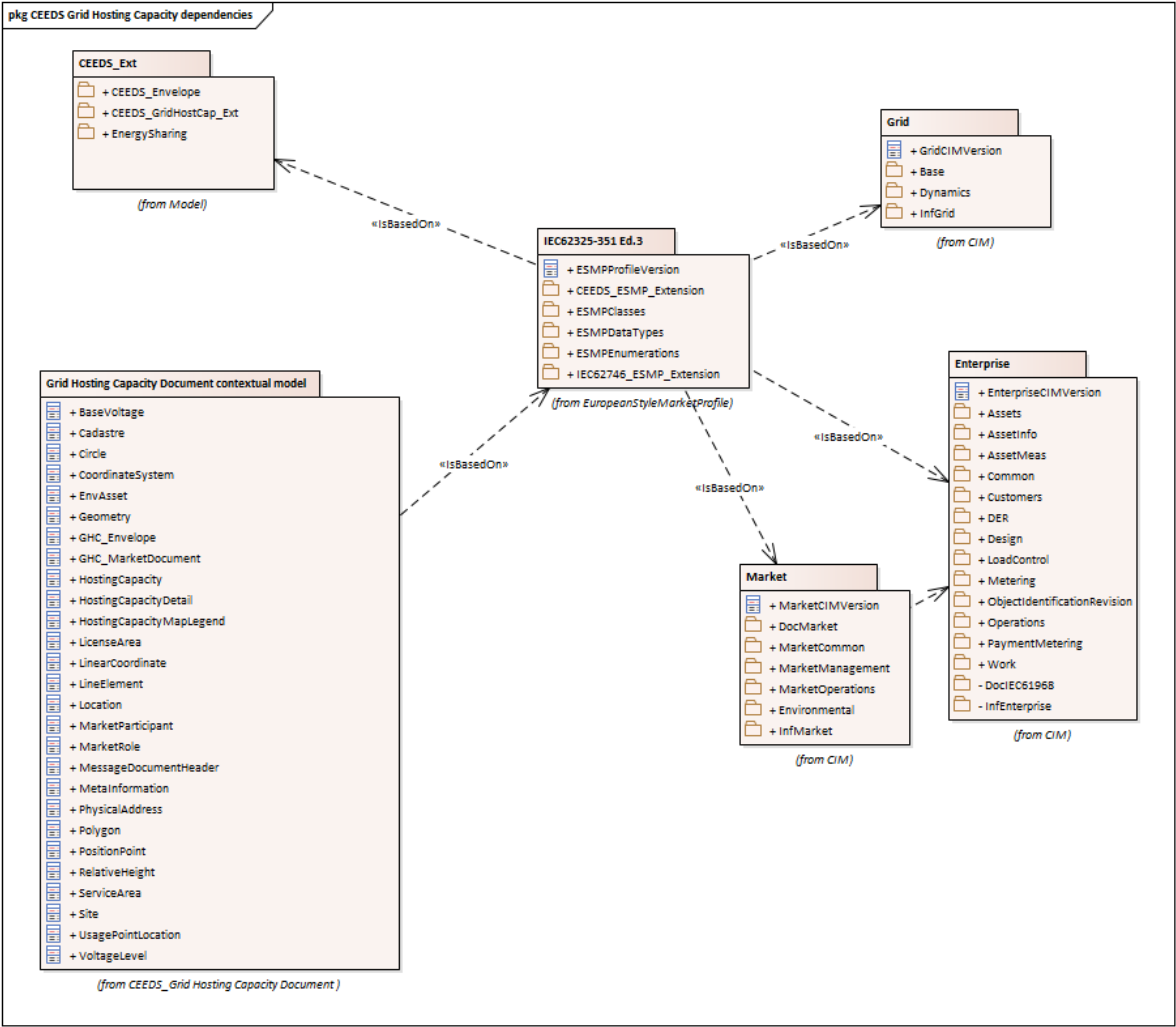


Figure !# - Grid Hosting Capacity Document contextual model

IsBasedOn relationships from the European style market profile

!# shows the traceability dependency of the classes used in this package towards the upper level.

Table !# - IsBasedOn dependency

| Name               | Complete IsBasedOn Path                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BaseVoltage        | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::BaseVoltage        |
| Cadastre           | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::Cadastre           |
| Circle             | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::Circle             |
| CoordinateSystem   | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::CoordinateSystem   |
| EnvAsset           | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::EnvAsset           |
| Geometry           | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::Geometry           |
| GHC_Envelope       | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::GHC_Envelope       |
| GHC_MarketDocument | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::GHC_MarketDocument |
| HostingCapacity    | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::HostingCapacity    |

|                          |                                                                                                                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HostingCapacityDetail    | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::HostingCapacityDetail    |
| HostingCapacityMapLegend | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::HostingCapacityMapLegend |
| LicenseArea              | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::LicenseArea              |
| LinearCoordinate         | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::LinearCoordinate         |
| LineElement              | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::LineElement              |
| Location                 | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::Location                 |
| MarketParticipant        | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::MarketParticipant        |
| MarketRole               | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::MarketRole               |
| MessageDocumentHeader    | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::MessageDocumentHeader    |
| MetalInformation         | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::MetalInformation         |
| PhysicalAddress          | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::PhysicalAddress          |
| Polygon                  | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::Polygon                  |
| PositionPoint            | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::PositionPoint            |
| RelativeHeight           | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::RelativeHeight           |
| ServiceArea              | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::ServiceArea              |
| Site                     | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::Site                     |
| UsagePointLocation       | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::UsagePointLocation       |
| VoltageLevel             | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document contextual model::VoltageLevel             |

##MXM##

## Grid Hosting Capacity Document assembly model

### Overview of the model

!# shows the model.

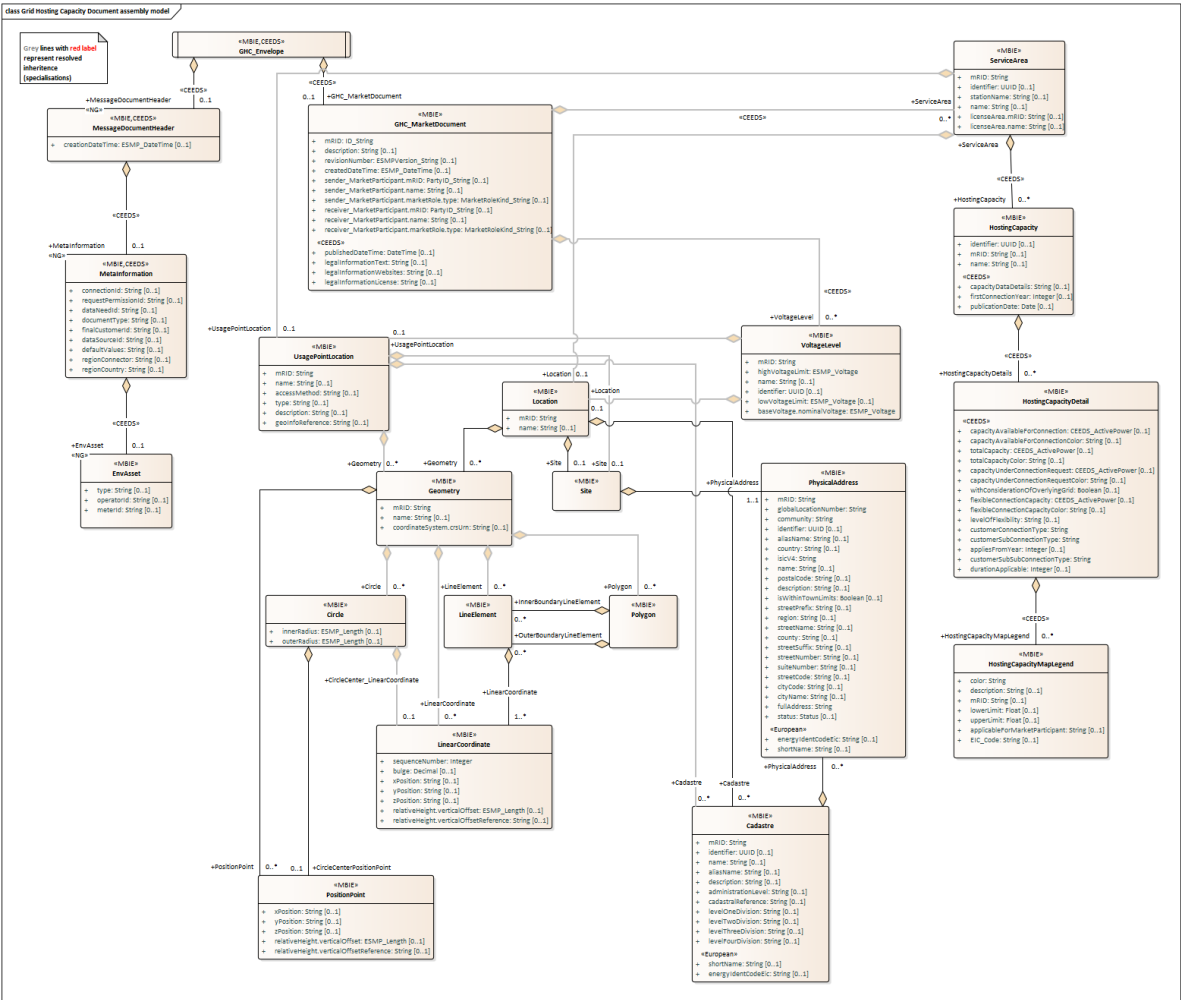


Figure !# - Grid Hosting Capacity Document assembly model

## IsBasedOn relationships from the European style market profile

# shows the traceability dependency of the classes used in this package towards the upper level.

### Table !# - IsBasedOn dependency

| Name                     | Complete IsBasedOn Path                                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cadastre                 | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::Cadastre                 |
| Circle                   | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::Circle                   |
| EnvAsset                 | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::EnvAsset                 |
| Geometry                 | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::Geometry                 |
| GHC_Envelope             | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::GHC_Envelope             |
| GHC_MarketDocument       | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::GHC_MarketDocument       |
| HostingCapacity          | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::HostingCapacity          |
| HostingCapacityDetail    | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::HostingCapacityDetail    |
| HostingCapacityMapLegend | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::HostingCapacityMapLegend |

|                       |                                                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LinearCoordinate      | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::LinearCoordinate      |
| LineElement           | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::LineElement           |
| Location              | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::Location              |
| MessageDocumentHeader | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::MessageDocumentHeader |
| MetaInformation       | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::MetaInformation       |
| PhysicalAddress       | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::PhysicalAddress       |
| Polygon               | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::Polygon               |
| PositionPoint         | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::PositionPoint         |
| ServiceArea           | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::ServiceArea           |
| Site                  | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::Site                  |
| UsagePointLocation    | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::UsagePointLocation    |
| VoltageLevel          | CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_Grid Hosting Capacity Document ::Grid Hosting Capacity Document assembly model::VoltageLevel          |

##MXM##

Detailed Grid Hosting Capacity Document assembly model

**GHC\_Envelope root class**

The envelope entity is a non-CIM object used in an energy data space environment to serve as a central top-level entity connecting the MarketDocumentHeader (energy data space specific context information block) with the regular CIM MarketDocument.

!# shows all association ends of GHC\_Envelope with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::GHC\_Envelope with other classes

| Order | mult.  | Class name / Role     | Description                                                                                                                                                                                                                         |
|-------|--------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [0..1] | MessageDocumentHeader | Association Based On :<br>MessageDocumentHeader Grid Hosting Capacity Document contextual model::GHC_Envelope.<br>[]<br>-----<br>Grid Hosting Capacity Document contextual model::MessageDocumentHeader.MessageDocumentHeader[0..1] |
| 1     | [0..1] | GHC_MarketDocument    | Association Based On :<br>GHC_MarketDocument Grid Hosting Capacity Document contextual model::GHC_Envelope.<br>[]<br>-----<br>Grid Hosting Capacity Document contextual model::GHC_MarketDocument.GHC_MarketDocument[0..1]          |

##MXM##

Cadastre

A singular complete record of all land parcels within some hierarchical level of administrative jurisdiction. The cadastre provides the ability to find the results of land parcel surveys typically used for land management and planning.

!# shows all attributes of Cadastre.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::Cadastre

| Order | mult.  | Attribute name / Attribute type | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|--------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..1] | mRID<br>String                  | Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended.<br>For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. |
| 1     | [0..1] | identifier<br>UUID              | A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.                                                                                                                                                                                                                                                                                                     |
| 2     | [0..1] | name<br>String                  | The name is any free human readable and possibly non unique text naming the object.                                                                                                                                                                                                                                                                                                                            |
| 3     | [0..1] | shortName<br>String             | The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.                                                                                                                                                                                                                                                                                          |
| 4     | [0..1] | aliasName<br>String             | The aliasName is free text human readable name of the object alternative to IdentifiedObject.name. It may be non unique and may not correlate to a naming hierarchy.<br>The attribute aliasName is retained because of backwards compatibility between CIM releases. It is however recommended to replace aliasName with the Name class as aliasName is planned for retirement at a future time.               |
| 5     | [0..1] | description<br>String           | The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.                                                                                                                                                                                                                                                               |
| 6     | [0..1] | administrationLevel<br>String   | Administrative level is referring to the organisation that is administrative authority for managing the a hierarchical division of a geographic area into land, property groups. This typically established by government that has the responsibility of managing governance, services, taxation, and legal jurisdiction, such as municipalities, townships, counties, or regions.                             |
| 7     | [0..1] | cadastralReference<br>String    | Local or national unique identification reference for the cadastral for land parcel, ensuring effective land management, ownership tracking, and legal compliance.                                                                                                                                                                                                                                             |
| 8     | [0..1] | energyIdentCodeEic<br>String    | The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. For details on EIC scheme please refer to ENTSO-E web site.                                                                                                                                                                                          |
| 9     | [0..1] | levelOneDivision<br>String      | Level One Division refers to the highest organizational or reference unit within a cadastral hierarchy, typically representing a primary division of land, such as a farm or property group, which can be further subdivided into smaller parcels.                                                                                                                                                             |
| 10    | [0..1] | levelTwoDivision<br>String      | Level Two Division refers to a subdivision of a Level One Parcel within a cadastral hierarchy, representing smaller, distinct land units typically used for individual ownership, usage, or development, such as specific plots or lots within a farm or property group.                                                                                                                                       |
| 11    | [0..1] | levelThreeDivision<br>String    | Level Three Division refers to a further subdivision of a Level Two Division within a cadastral hierarchy, typically used to represent specialized land units such as                                                                                                                                                                                                                                          |

|                                                                                                           |        |                                        |                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------|--------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                           |        |                                        | leased parcels, easements, or other secondary divisions associated with distinct legal or usage rights.                                                                                                                                                             |
| 12                                                                                                        | [0..1] | levelFourDivision<br><br>String        | Level Four Division refers to a further subdivision of a Level Three Division within a cadastral hierarchy, typically used to represent specific units or sections within a shared property or structure, such as apartments, condominiums, or co-owned facilities. |
| ##MXM##                                                                                                   |        |                                        |                                                                                                                                                                                                                                                                     |
| !# shows all association ends of Cadastre with other classes.                                             |        |                                        |                                                                                                                                                                                                                                                                     |
| Table !# - Association ends of Grid Hosting Capacity Document assembly model::Cadastre with other classes |        |                                        |                                                                                                                                                                                                                                                                     |
| Order                                                                                                     | mult.  | Class name / Role                      | Description                                                                                                                                                                                                                                                         |
| 13                                                                                                        | [0..*] | PhysicalAddress<br><br>PhysicalAddress | Physical addresses encompassed within a given cadastre.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::Cadastre.[]<br>-----<br>Grid Hosting Capacity Document contextual model::PhysicalAddress.PhysicalAddress[0..*]                |
| ##MXM##                                                                                                   |        |                                        |                                                                                                                                                                                                                                                                     |

|                                                                                                                                                                                                                                                                                                           |        |                                                       |                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Circle</b>                                                                                                                                                                                                                                                                                             |        |                                                       |                                                                                                                                                                                                                                                                                                      |
| A circular area in space, described by a center point and radius. Any circle object must have a center and an outer radius and may have an inner radius. If an inner radius is specified, the area within the inner radius is to be excluded from the area within the outer radius, forming a ring shape. |        |                                                       |                                                                                                                                                                                                                                                                                                      |
| !# shows all attributes of Circle.                                                                                                                                                                                                                                                                        |        |                                                       |                                                                                                                                                                                                                                                                                                      |
| Table !# - Attributes of Grid Hosting Capacity Document assembly model::Circle                                                                                                                                                                                                                            |        |                                                       |                                                                                                                                                                                                                                                                                                      |
| Order                                                                                                                                                                                                                                                                                                     | mult.  | Attribute name / Description                          | Attribute type                                                                                                                                                                                                                                                                                       |
| 0                                                                                                                                                                                                                                                                                                         | [0..1] | innerRadius                                           | A circle may have an inner radius. If specified the area described by the inner radius is to be excluded from the area within the outer radius, forming a ring shape.                                                                                                                                |
|                                                                                                                                                                                                                                                                                                           |        | ESMP_Length                                           |                                                                                                                                                                                                                                                                                                      |
| 1                                                                                                                                                                                                                                                                                                         | [0..1] | outerRadius                                           | Every circle must have an outer radius which describes the outer limit of the circular area.                                                                                                                                                                                                         |
|                                                                                                                                                                                                                                                                                                           |        | ESMP_Length                                           |                                                                                                                                                                                                                                                                                                      |
| ##MXM##                                                                                                                                                                                                                                                                                                   |        |                                                       |                                                                                                                                                                                                                                                                                                      |
| !# shows all association ends of Circle with other classes.                                                                                                                                                                                                                                               |        |                                                       |                                                                                                                                                                                                                                                                                                      |
| Table !# - Association ends of Grid Hosting Capacity Document assembly model::Circle with other classes                                                                                                                                                                                                   |        |                                                       |                                                                                                                                                                                                                                                                                                      |
| Order                                                                                                                                                                                                                                                                                                     | mult.  | Class name / Role                                     | Description                                                                                                                                                                                                                                                                                          |
| 2                                                                                                                                                                                                                                                                                                         | [0..1] | PositionPoint<br><br>CircleCenterPositionPoint        | A point representing the center of a circular area. Any circle object must have one CircleCenter.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::Circle.[]<br>-----<br>Grid Hosting Capacity Document contextual model::PositionPoint.CircleCenterPositionPoint[0..1] |
| 3                                                                                                                                                                                                                                                                                                         | [0..1] | LinearCoordinate<br><br>CircleCenter_LinearCoordinate | A point representing the center of a circular area. Any circle object must have one CircleCenter.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::Circle.[]                                                                                                            |

-----  
Grid Hosting Capacity Document contextual  
model::LinearCoordinate.CircleCenter\_LinearCoordinate[0..1]

##MXM##

EnvAsset

EnvAsset is an energy data space specific attribute referring to the Asset from which data is being requested

!# shows all attributes of EnvAsset.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::EnvAsset

| Order | mult.  | Attribute name / Attribute type | Description                                                                                                    |
|-------|--------|---------------------------------|----------------------------------------------------------------------------------------------------------------|
| 0     | [0..1] | type<br>String                  | type is an energy data space specific attribute that specifies type of asset refered to.                       |
| 1     | [0..1] | operatorId<br>String            | operatorId is an energy data space specific attribute that specifies the id operator of the asset refered to.  |
| 2     | [0..1] | meterId<br>String               | meterId is an energy data space specific attribute that specifies the id of the meter that measures the asset. |

##MXM##

Geometry

Geo-spatial representation of a physical location. A location may be represented as a Point, Line, Polygon, or Circle.

!# shows all attributes of Geometry.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::Geometry

| Order | mult.  | Attribute name / Attribute type   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|--------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..1] | mRID<br>String                    | Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended.<br>For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.                                                                                                                                                                                                              |
| 1     | [0..1] | name<br>String                    | The name is any free human readable and possibly non unique text naming the object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2     | [0..1] | coordinateSystem.crsUrn<br>String | A Uniform Resource Name (URN) for the coordinate reference system (crs) used to define 'Location.PositionPoints'.<br>An example would be the European Petroleum Survey Group (EPSG) code for a coordinate reference system, defined in URN under the Open Geospatial Consortium (OGC) namespace as:<br>urn:ogc:def:uom:EPSG::XXXX, where XXXX is an EPSG code (a full list of codes can be found at the EPSG Registry web site <a href="http://www.epsg-registry.org/">http://www.epsg-registry.org/</a> ). To define the coordinate system as being WGS84 (latitude, longitude) using an EPSG OGC, this attribute would be |



urn:ogc:def:uom:EPSG::4236.  
A profile should limit this code to a set of allowed URNs agreed to by all sending and receiving parties.  
--- The coordinate system used for a given geometry.

##MXM##

!# shows all association ends of Geometry with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::Geometry with other classes

| Order | mult.  | Class name / Role                    | Description                                                                                                                                                                                 |
|-------|--------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | [0..*] | PositionPoint<br>PositionPoint       | Association Based On :<br>Grid Hosting Capacity Document contextual model::Geometry.[]<br>-----<br>Grid Hosting Capacity Document contextual model::PositionPoint.PositionPoint[0..*]       |
| 4     | [0..*] | LineElement<br>LineElement           | Association Based On :<br>Grid Hosting Capacity Document contextual model::Geometry.[]<br>-----<br>Grid Hosting Capacity Document contextual model::LineElement.LineElement[0..*]           |
| 5     | [0..*] | Polygon<br>Polygon                   | Association Based On :<br>Grid Hosting Capacity Document contextual model::Geometry.[]<br>-----<br>Grid Hosting Capacity Document contextual model::Polygon.Polygon[0..*]                   |
| 6     | [0..*] | Circle<br>Circle                     | Association Based On :<br>Grid Hosting Capacity Document contextual model::Geometry.[]<br>-----<br>Grid Hosting Capacity Document contextual model::Circle.Circle[0..*]                     |
| 7     | [0..*] | LinearCoordinate<br>LinearCoordinate | Association Based On :<br>Grid Hosting Capacity Document contextual model::Geometry.[]<br>-----<br>Grid Hosting Capacity Document contextual model::LinearCoordinate.LinearCoordinate[0..*] |

##MXM##

GHC\_MarketDocument

An electronic document containing the information necessary to satisfy the requirements of a given business process.

!# shows all attributes of GHC\_MarketDocument.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::GHC\_MarketDocument

| Order | mult.  | Attribute name / Attribute type | Description                                                                                                                                                                                                                                                               |
|-------|--------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..1] | mRID<br><br>ID_String           | The unique identification of the document being exchanged within a business process flow.<br><br>In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides an identification in the context of a business exchange such |

|    |                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                                                          | as document identification, ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|    |                                                                          | Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.                                                                                                                                                                                                                                                                                                                                        |
| 1  | [0..1] description<br>String                                             | The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2  | [0..1] publishedDateTime<br>DateTime                                     | The date time the legal information is published                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3  | [0..1] revisionNumber<br>ESMPVersion_String                              | The identification of the version that distinguishes one evolution of a document from another.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4  | [0..1] legalInformationText<br>String                                    | Text with the required legal information from the information provider (Connecting System operator=CSO)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5  | [0..1] legalInformationWebsites<br>String                                | URL pointing to the required legal information from the information provider (Connecting System operator=CSO)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6  | [0..1] legalInformationLicense<br>String                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7  | [0..1] createdDateTime<br>ESMP_DateTime                                  | The date and time of the creation of the document.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 10 | [0..1] sender_MarketParticipant.mRID<br>PartyID_String                   | <p>The identification of a party in the energy market.</p> <p>In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification.</p> <p>Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.</p> |
| 11 | [0..1] sender_MarketParticipant.name<br>String                           | The name is any free human readable and possibly non unique text naming the object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 12 | [0..1] sender_MarketParticipant.marketRole.type<br>MarketRoleKind_String | The identification of the role played by a market player.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 13 | [0..1] receiver_MarketParticipant.mRID<br>PartyID_String                 | <p>The identification of a party in the energy market.</p> <p>In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification.</p> <p>Master resource identifier issued by a model authority.</p>                                                                                                                                                                                                                                                                                                |

|                                                                                                                     |        |                                                                     |                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                     |        |                                                                     | The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. |
| 14                                                                                                                  | [0..1] | receiver_MarketParticipant.name<br>String                           | The name is any free human readable and possibly non unique text naming the object.                                                                                                                                                                                                            |
| 15                                                                                                                  | [0..1] | receiver_MarketParticipant.marketRole.type<br>MarketRoleKind_String | The identification of the role played by a market player.                                                                                                                                                                                                                                      |
| ##MXM##                                                                                                             |        |                                                                     |                                                                                                                                                                                                                                                                                                |
| !# shows all association ends of GHC_MarketDocument with other classes.                                             |        |                                                                     |                                                                                                                                                                                                                                                                                                |
| Table !# - Association ends of Grid Hosting Capacity Document assembly model::GHC_MarketDocument with other classes |        |                                                                     |                                                                                                                                                                                                                                                                                                |
| Order                                                                                                               | mult.  | Class name / Role                                                   | Description                                                                                                                                                                                                                                                                                    |
| 8                                                                                                                   | [0..*] | VoltageLevel<br>VoltageLevel                                        | Association Based On :<br>Grid Hosting Capacity Document contextual model::GHC_MarketDocument.[]<br>-----<br>Grid Hosting Capacity Document contextual model::VoltageLevel.VoltageLevel[0..*]                                                                                                  |
| 9                                                                                                                   | [0..*] | ServiceArea<br>ServiceArea                                          | Association Based On :<br>Grid Hosting Capacity Document contextual model::GHC_MarketDocument.[]<br>-----<br>Grid Hosting Capacity Document contextual model::ServiceArea.ServiceArea[0..*]                                                                                                    |
| ##MXM##                                                                                                             |        |                                                                     |                                                                                                                                                                                                                                                                                                |

## HostingCapacity

A structure for a set of information on grid hosting capacity.

!# shows all attributes of HostingCapacity.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::HostingCapacity

|       |        |                                 |                                                                                                                                                                                                                                                                                                        |
|-------|--------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Order | mult.  | Attribute name / Attribute type | Description                                                                                                                                                                                                                                                                                            |
| 0     | [0..1] | identifier<br>UUID              | A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.                                                                                                                                                                                             |
| 1     | [0..1] | capacityDataDetails<br>String   | Informational field for further information on <ul style="list-style-type: none"><li>timing (e.g. is time for capacity available for connection a <b>snapshot</b> or is it the time when connection can be <b>performed</b>)</li><li>description of the SO description of the data provided,</li></ul> |

- description of the meaning of the different colours

|   |                            |                                                                                                                                                                                                                                                                                                                                                                                                                |
|---|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | [0..1] mRID                | Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended.<br>For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. |
|   | String                     |                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3 | [0..1] firstConnectionYear | The year that the capacity is applicable.                                                                                                                                                                                                                                                                                                                                                                      |
|   | Integer                    |                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4 | [0..1] name                | The name is any free human readable and possibly non unique text naming the object.                                                                                                                                                                                                                                                                                                                            |
|   | String                     |                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5 | [0..1] publicationDate     | The date when the data was produced or when it can be available                                                                                                                                                                                                                                                                                                                                                |
|   | Date                       |                                                                                                                                                                                                                                                                                                                                                                                                                |

##MXM##

!# shows all association ends of HostingCapacity with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::HostingCapacity with other classes

| Order | mult.  | Class name / Role     | Description                                                                                                     |
|-------|--------|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| 6     | [0..*] | HostingCapacityDetail | Association Based On :<br>HostingCapacityDetails                                                                |
|       |        |                       | Grid Hosting Capacity Document contextual model::HostingCapacity.[]                                             |
|       |        |                       | -----<br>Grid Hosting Capacity Document contextual<br>model::HostingCapacityDetail.HostingCapacityDetails[0..*] |

##MXM##

HostingCapacityDetail

!# shows all attributes of HostingCapacityDetail.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::HostingCapacityDetail

| Order | mult.  | Attribute name / Attribute type     | Description |
|-------|--------|-------------------------------------|-------------|
| 0     | [0..1] | capacityAvailableForConnection      |             |
|       |        | CEEDS_ActivePower                   |             |
| 1     | [0..1] | capacityAvailableForConnectionColor |             |
|       |        | String                              |             |
| 2     | [0..1] | totalCapacity                       |             |
|       |        | CEEDS_ActivePower                   |             |
| 3     | [0..1] | totalCapacityColor                  |             |
|       |        | String                              |             |
| 4     | [0..1] | capacityUnderConnectionRequest      |             |
|       |        | CEEDS_ActivePower                   |             |
| 5     | [0..1] | capacityUnderConnectionRequestColor |             |

|    |                                        |
|----|----------------------------------------|
|    | String                                 |
| 6  | [0..1]withConsiderationOfOverlyingGrid |
|    | Boolean                                |
| 7  | [0..1]flexibleConnectionCapacity       |
|    | CEEDS_ActivePower                      |
| 8  | [0..1]flexibleConnectionCapacityColor  |
|    | String                                 |
| 9  | [0..1]levelOfFlexibility               |
|    | String                                 |
| 10 | [1..1]customerConnectionType           |
|    | String                                 |
| 11 | [1..1]customerSubConnectionType        |
|    | String                                 |
| 12 | [0..1]appliesFromYear                  |
|    | Integer                                |
| 13 | [1..1]customerSubSubConnectionType     |
|    | String                                 |
| 14 | [0..1]durationApplicable               |
|    | Integer                                |

##MXM##

!# shows all association ends of HostingCapacityDetail with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::HostingCapacityDetail with other classes

| Order | mult.  | Class name / Role                                    | Description                                                                                                                                                                                                                  |
|-------|--------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | [0..*] | HostingCapacityMapLegend<br>HostingCapacityMapLegend | Association Based On :<br>Grid Hosting Capacity Document contextual<br>model::HostingCapacityDetail.<br>-----<br>Grid Hosting Capacity Document contextual<br>model::HostingCapacityMapLegend.HostingCapacityMapLegend[0..*] |

##MXM##

HostingCapacityMapLegend

!# shows all attributes of HostingCapacityMapLegend.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::HostingCapacityMapLegend

| Order | mult.  | Attribute name / Attribute type | Description                                                                                                                                      |
|-------|--------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..1] | color                           |                                                                                                                                                  |
|       |        | String                          |                                                                                                                                                  |
| 1     | [0..1] | description                     | The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy. |
|       |        | String                          |                                                                                                                                                  |

|         |                                      |                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2       | [0..1]mRID                           | Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. |
|         | String                               |                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3       | [0..1]lowerLimit                     |                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | Float                                |                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4       | [0..1]upperLimit                     |                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | Float                                |                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5       | [0..1]applicableForMarketParticipant |                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | String                               |                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6       | [0..1]EIC_Code                       |                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | String                               |                                                                                                                                                                                                                                                                                                                                                                                                             |
| ##MXM## |                                      |                                                                                                                                                                                                                                                                                                                                                                                                             |

LinearCoordinate

Specifies a point on a line segment, including a sequence number and bulge. To visualize the line segment, each linear coordinate is to be connected in the order specified by the sequence attribute. Each linear coordinate must have a sequence value.

!# shows all attributes of LinearCoordinate.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::LinearCoordinate

| Order | mult.  | Attribute name / Attribute type        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|--------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..1] | sequenceNumber                         | The order in which linear coordinates are to be connected to form the line described. Sequence is required for all linear coordinates.                                                                                                                                                                                                                                                                                                            |
|       |        | Integer                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1     | [0..1] | bulge                                  | Bulge is a factor that describes a curved line segment between two points. bulge=0 is a straight line: bulge=1 is a curve along the edge of a circle. A bulge can be associated only with the second point of a line segment. A negative bulge factor causes the line to bulge to the left as viewed from the first point to the second; a positive bulge causes the line to be curved to the right as viewed from the first point to the second. |
|       |        | Decimal                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2     | [0..1] | xPosition                              | X axis position.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|       |        | String                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3     | [0..1] | yPosition                              | Y axis position.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|       |        | String                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4     | [0..1] | zPosition                              | (if applicable) Z axis position.                                                                                                                                                                                                                                                                                                                                                                                                                  |
|       |        | String                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5     | [0..1] | relativeHeight.verticalOffset          | Distance above (positive value) or below (negative value) the specified vertical offset reference.                                                                                                                                                                                                                                                                                                                                                |
|       |        | ESMP_Length                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6     | [0..1] | relativeHeight.verticalOffsetReference | The height from which vertical offset is measured. Examples include "Ground Level" , "Pole Top" , "Tower                                                                                                                                                                                                                                                                                                                                          |

|         |              |
|---------|--------------|
| String  | Topâ€ , etc. |
| ##MXM## |              |

LineElement

A geometric line segment, described by 2 or more coordinate tuples with linear interpolation between them. A line must contain at least 2 linear coordinates.

!# shows all association ends of LineElement with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::LineElement with other classes

| Order | mult.  | Class name / Role | Description                                                                                                                                                                                                                                                                             |
|-------|--------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..*] | LinearCoordinate  | The linear coordinates which define a line with one or more segments.<br>Association Based On :<br>LinearCoordinateGrid Hosting Capacity Document contextual model::LineElement.[]<br>-----<br>Grid Hosting Capacity Document contextual model::LinearCoordinate.LinearCoordinate[1..*] |

##MXM##

Location

The place, scene, or point of something where someone or something has been, is, and/or will be at a given moment in time. It can be defined with one or more position points (coordinates) in a given coordinate system.

!# shows all attributes of Location.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::Location

| Order | mult.  | Attribute name / Attribute type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|--------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..1] | mRID                            | The unique identification of a location.<br><br>String<br>In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification, location, etc.<br><br>Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context.<br>Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this.<br>For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. |
| 1     | [0..1] | name                            | The name is any free human readable and possibly non unique text naming the object.<br><br>String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

##MXM##

!# shows all association ends of Location with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::Location with other classes

| Order | mult.  | Class name / Role    | Description                                                                                                                                                                                                       |
|-------|--------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2     | [0..*] | Geometry<br>Geometry | The geometry for a given location.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::Location.[]<br>-----<br>Grid Hosting Capacity Document contextual model::Geometry.Geometry[0..*] |
| 3     | [0..1] | Site<br>Site         | Association Based On :<br>Grid Hosting Capacity Document contextual model::Location.[]<br>-----<br>Grid Hosting Capacity Document contextual model::Site.Site[0..1]                                               |
| 4     | [0..*] | Cadastre<br>Cadastre | Cadastre encompassing a location.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::Location.[]<br>-----<br>Grid Hosting Capacity Document contextual model::Cadastre.Cadastre[0..*]  |

##MXM##

MessageDocumentHeader

MessageDocumentHeader is an energy data space specific entity with the creationDateTime of this document.

!# shows all attributes of MessageDocumentHeader.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::MessageDocumentHeader

| Order | mult.  | Attribute name / Attribute type   | Description                                             |
|-------|--------|-----------------------------------|---------------------------------------------------------|
| 0     | [0..1] | creationDateTime<br>ESMP_DateTime | The date and time the MessageDocumenHeader was created. |

##MXM##

!# shows all association ends of MessageDocumentHeader with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::MessageDocumentHeader with other classes

| Order | mult.  | Class name / Role                  | Description                                                                                                                                                                                            |
|-------|--------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | [0..1] | MetaInformation<br>MetaInformation | Association Based On :<br>Grid Hosting Capacity Document contextual model::MessageDocumentHeader.[]<br>-----<br>Grid Hosting Capacity Document contextual model::MetaInformation.MetaInformation[0..1] |

##MXM##

MetaInformation

MetaInformation is an energy data space specific entity referring to the information concerning the technical and functional context of the MarketDocument.



!# shows all attributes of MetaInformation.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::MetaInformation

| Order | mult.  | Attribute name / Attribute type | Description                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|--------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [0..1] | connectionId<br>String          | connectionId is an energy data space specific attribute referring to the established connection used for the communication of this document.                                                                                                                                                                                                                                                              |
| 1     | [0..1] | requestPermissionId<br>String   | The requestPermissionId is energy data space specific identification attribute holding an unique identifier issued by the energy data space to allow for tracing of the permission through serveral message exchanges.<br>Note: the requestPermissionId is NOT the identifier issued by the Identity Service Provider as specified in the EU Implementing Act on access to Metering and Consumption Data. |
| 2     | [0..1] | dataNeedId<br>String            | dataNeedId is an energy data space specific attribute holding the type of data need requested, like "last 24 hours at 1 hour interval" .                                                                                                                                                                                                                                                                  |
| 3     | [0..1] | documentType<br>String          | documentType is an energy data space specific attribute referring to the type of document, comparable to the MessageTypeList in the CIM part of the message.<br>Note:<br>This is applicable if you have the need to know the content without having to open the complete document. DocumentType might contain messages not (yet) part of the MessageTypeList.                                             |
| 4     | [0..1] | finalCustomerId<br>String       | The finalCustomerId is an energy data space specific attribute identifying the actor doing the request for data or data sharing.                                                                                                                                                                                                                                                                          |
| 5     | [0..1] | dataSourceId<br>String          | dataSourceId is an energy data space specific attribute referring to the datasource from which the requested data is being queried.                                                                                                                                                                                                                                                                       |
| 6     | [0..1] | defaultValues<br>String         | defaultValues is a energy data space specific attribute to hold relevant default values. The format to use can freely be chosen based on the implementation. Most commonly you would expect a semi-colon separated name value pair, like: "minPower=value; maxPower=value; avgPower=value".                                                                                                               |
| 7     | [0..1] | regionConnector<br>String       | regionConnector is an energy data space specific attribute that specifies the regional connector being used to retrieve the data requested.                                                                                                                                                                                                                                                               |
| 8     | [0..1] | regionCountry<br>String         | regionCountry is an energy data space specific attribute that specifies the country where the regional connector is located.                                                                                                                                                                                                                                                                              |

##MXM##

!# shows all association ends of MetaInformation with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::MetaInformation with other classes

| Order | mult.  | Class name / Role    | Description                                                                                                                                                                        |
|-------|--------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9     | [0..1] | EnvAsset<br>EnvAsset | Association Based On :<br>Grid Hosting Capacity Document contextual model::MetaInformation.[]<br>-----<br>Grid Hosting Capacity Document contextual model::EnvAsset.EnvAsset[0..1] |

##MXM##

PhysicalAddress

The structured geographic and administrative context of a physical place where an organisation, person, or activity is located.

PhysicalAddress is used for precisely identifying, classifying, and locating entities for regulatory, logistical, or informational purposes.

!# shows all attributes of PhysicalAddress.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::PhysicalAddress

| Order | mult. Attribute name / Attribute type | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..1] mRID<br>String                 | Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended.<br>For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. |
| 1     | [1..1] globalLocationNumber<br>String | Global Location Number (GLN, sometimes also referred to as International Location Number or ILN) of the respective organization, person, or place. The GLN is a 13-digit number used to identify parties and physical locations.                                                                                                                                                                               |
| 2     | [0..1] energyIdentCodeEic<br>String   | The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. For details on EIC scheme please refer to ENTSO-E web site.                                                                                                                                                                                          |
| 3     | [1..1] community<br>String            | Community in which the locality is, and which is in the municipality, city or town.. For example, Anderston, Glasgow City, Scotland or another appropriate fourth-level administrative division                                                                                                                                                                                                                |
| 4     | [0..1] identifier<br>UUID             | A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.                                                                                                                                                                                                                                                                                                     |
| 5     | [0..1] aliasName<br>String            | The aliasName is free text human readable name of the object alternative to IdentifiedObject.name. It may be non unique and may not correlate to a naming hierarchy.<br>The attribute aliasName is retained because of backwards compatibility between CIM releases. It is however recommended to replace aliasName with the Name class as aliasName is planned for retirement at a future time.               |
| 6     | [0..1] country<br>String              | Name of the country in which the address is located. It is recommended to use either the 3-letter ISO 3166-1 alpha 3 code such as "SGP," or the full country name such as "Singapore."                                                                                                                                                                                                                         |
| 7     | [1..1] isicV4<br>String               | International Standard of Industrial Classification of All Economic Activities (ISIC), Revision 4 code for a particular organization, business person, or place.                                                                                                                                                                                                                                               |
| 8     | [0..1] name<br>String                 | The name is any free human readable and possibly non unique text naming the object.                                                                                                                                                                                                                                                                                                                            |
| 9     | [0..1] postalCode<br>String           | Postal code for the address.                                                                                                                                                                                                                                                                                                                                                                                   |
| 10    | [0..1] shortName<br>String            | The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.                                                                                                                                                                                                                                                                                          |
| 11    | [0..1] description<br>String          | The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.                                                                                                                                                                                                                                                               |
| 12    | [0..1] isWithinTownLimits<br>Boolean  | If TRUE, this street is within the legal geographical boundaries of the specified town (default).                                                                                                                                                                                                                                                                                                              |
| 13    | [0..1] streetPrefix<br>String         | Prefix to the street name. For example: North, South, East, West.                                                                                                                                                                                                                                                                                                                                              |

|         |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14      | [0..1]region       | Name of the first-level Administrative division of a country in which the address is located. For example, California, Quebec, and Queensland are states and provinces of the United States of America, Canada, and Australia, respectively.                                                                                                                                                                                                                                                                  |
|         | String             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 15      | [0..1]streetName   | Name of the street.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|         | String             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 16      | [0..1]county       | Name of the second-level Administrative division of a country in which the address is located. The term used for the second-level Administrative division in a county varies from country to country and between first-level Administrative divisions within a country. For example, California has counties, Louisiana has parishes, British Columbia, Canada has regional districts, and Queensland Australia has shires and regions. Counties are usually made up of cities, towns, and rural populations. |
|         | String             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 17      | [0..1]streetSuffix | Suffix to the street name. For example: North, South, East, West.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | String             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 18      | [0..1]streetNumber | Designator of the specific location on the street.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|         | String             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 19      | [0..1]suiteNumber  | Number of the apartment or suite.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | String             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 20      | [0..1]streetCode   | (if applicable) Utilities often make use of external reference systems, such as those of the town-planner's department or surveyor general's mapping system, that allocate global reference codes to streets.                                                                                                                                                                                                                                                                                                 |
|         | String             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 21      | [0..1]cityCode     | City code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|         | String             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 22      | [0..1]cityName     | Name of the city.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | String             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 23      | [1..1]fullAddress  | A concatenated string specifying the complete address in a single string which may include delimiters.                                                                                                                                                                                                                                                                                                                                                                                                        |
|         | String             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 24      | [0..1]status       | Current status of the address e.g. pending, approved                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|         | Status             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ##MXM## |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Polygon

One or more self-closing line segments representing a polygonal area. In order to be self-closing, the first and last linear coordinates of each line segment must be identical. A polygon must have one outer boundary and may have one or more inner boundaries. If an inner boundary is specified, the area within the inner boundary is to be excluded from the area within the outer boundary. Each line description within a polygon (both outer and inner boundaries) must contain at least 4 linear coordinates.

!# shows all association ends of Polygon with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::Polygon with other classes

| Order | mult.  | Class name / Role        | Description                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|--------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [0..*] | LineElement              | Optional for all polygons. If specified, an inner boundary must contain at least 4 linear coordinates. The first and last coordinates in the sequence must be identical. The area described by an inner boundary is to be excluded from the area described by the outer boundary in order to complete the description of a complex polygon. The entire area described by an inner boundary should |
|       |        | InnerBoundaryLineElement |                                                                                                                                                                                                                                                                                                                                                                                                   |

|   |        |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|--------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |        |                                             | fall within the area described by the outer boundary.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::Polygon.[]<br>-----<br>Grid Hosting Capacity Document contextual<br>model::LineElement.InnerBoundaryLineElement[0..*]                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1 | [0..*] | LineElement<br><br>OuterBoundaryLineElement | One or more self-closing line segments representing a polygonal area. In order to be self-closing, the first and last linear coordinates of each line segment must be identical. A polygon must have one outer boundary and may have one or more inner boundaries. If an inner boundary is specified, the area within the inner boundary is to be excluded from the area within the outer boundary. Each line description within a polygon (both outer and inner boundaries) must contain at least 4 linear coordinates.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::Polygon.[]<br>-----<br>Grid Hosting Capacity Document contextual<br>model::LineElement.OuterBoundaryLineElement[0..*] |

##MXM##

PositionPoint

Set of spatial coordinates that determine a point, defined in the coordinate system specified in 'Location.CoordinateSystem'. Use a single position point instance to describe a point-oriented location. Use a sequence of position points to describe a line-oriented object (physical location of non-point oriented objects like cables or lines), or area of an object (like a substation or a geographical zone - in this case, have first and last position point with the same values).

!# shows all attributes of PositionPoint.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::PositionPoint

| Order | mult.  | Attribute name / Attribute type                      | Description                                                                                                          |
|-------|--------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 0     | [0..1] | xPosition<br><br>String                              | X axis position.                                                                                                     |
| 1     | [0..1] | yPosition<br><br>String                              | Y axis position.                                                                                                     |
| 2     | [0..1] | zPosition<br><br>String                              | (if applicable) Z axis position.                                                                                     |
| 3     | [0..1] | relativeHeight.verticalOffset<br><br>ESMP_Length     | Distance above (positive value) or below (negative value) the specified vertical offset reference.                   |
| 4     | [0..1] | relativeHeight.verticalOffsetReference<br><br>String | The height from which vertical offset is measured. Examples include "Ground Level" , "Pole Top" , "Tower Top" , etc. |

##MXM##

ServiceArea

!# shows all attributes of ServiceArea.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::ServiceArea

| Order | mult.  | Attribute name / Attribute type | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|--------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..1] | mRID<br>String                  | Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended.<br>For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. |
| 1     | [0..1] | identifier<br>UUID              | A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.                                                                                                                                                                                                                                                                                                     |
| 2     | [0..1] | stationName<br>String           | The human readable name of the station.                                                                                                                                                                                                                                                                                                                                                                        |
| 3     | [0..1] | name<br>String                  | The name is any free human readable and possibly non unique text naming the object.                                                                                                                                                                                                                                                                                                                            |
| 6     | [0..1] | licenseArea.mRID<br>String      | Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended.<br>For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. |
| 7     | [0..1] | licenseArea.name<br>String      | The name is any free human readable and possibly non unique text naming the object.                                                                                                                                                                                                                                                                                                                            |

##MXM##

!# shows all association ends of ServiceArea with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::ServiceArea with other classes

| Order | mult.  | Class name / Role                        | Description                                                                                                                                                                                                                                   |
|-------|--------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4     | [0..1] | Location<br>Location                     | Location of this power system resource.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::Location.Location[0..1]<br>-----<br>Grid Hosting Capacity Document contextual model::ServiceArea.[]                     |
| 5     | [0..1] | UsagePointLocation<br>UsagePointLocation | Location of this power system resource.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::ServiceArea.[]<br>-----<br>Grid Hosting Capacity Document contextual model::UsagePointLocation.UsagePointLocation[0..1] |
| 8     | [0..*] | HostingCapacity<br>HostingCapacity       | Association Based On :<br>Grid Hosting Capacity Document contextual model::ServiceArea.ServiceArea[]<br>-----<br>Grid Hosting Capacity Document contextual model::HostingCapacity.HostingCapacity[0..*]                                       |

##MXM##

Area of ground on which work can be executed, organisation can be hosted, and building (including substation and power line) can be located.

!# shows all association ends of Site with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::Site with other classes

| Order | mult.  | Class name / Role | Description                                                                                                                                                                                           |
|-------|--------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..1] | PhysicalAddress   | Association Based On :<br>PhysicalAddress Grid Hosting Capacity Document contextual model::Site.[]<br>-----<br>Grid Hosting Capacity Document contextual model::PhysicalAddress.PhysicalAddress[1..1] |

##MXM##

UsagePointLocation

Location of an individual usage point.

!# shows all attributes of UsagePointLocation.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::UsagePointLocation

| Order | mult.  | Attribute name / Attribute type                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|--------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | [1..1] | mRID<br><br>String                             | The unique identification of a location.<br><br>In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification, location, etc.<br><br>Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context.<br>Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this.<br>For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. |
| 1     | [0..1] | name<br><br>String                             | The name is any free human readable and possibly non unique text naming the object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2     | [0..1] | accessMethod<br><br>String                     | Method for the service person to access this usage point location. For example, a description of where to obtain a key if the facility is unmanned and secured.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3     | [0..1] | type<br><br>String                             | Classification by utility's corporate standards and practices, relative to the location itself (e.g., geographical, functional accounting, etc., not a given property that happens to exist at that location).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4     | [0..1] | description<br><br>String                      | The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5     | [0..1] | geoInfoReference (if applicable)<br><br>String | Reference to geographical information source, often external to the utility.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

##MXM##

!# shows all association ends of UsagePointLocation with other classes.

Table !# - Association ends of Grid Hosting Capacity Document assembly model::UsagePointLocation with other classes

| Order   | mult.  | Class name / Role        | Description                                                                                                                                                                                                                 |
|---------|--------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6       | [0..*] | Geometry<br><br>Geometry | The geometry for a given location.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::Geometry.Geometry[0..*]<br>-----<br>Grid Hosting Capacity Document contextual model::UsagePointLocation.[] |
| 7       | [0..1] | Site<br><br>Site         | Association Based On :<br>Grid Hosting Capacity Document contextual model::Site.Site[0..1]<br>-----<br>Grid Hosting Capacity Document contextual model::UsagePointLocation.[]                                               |
| 8       | [0..*] | Cadastre<br><br>Cadastre | Cadastre encompassing a location.<br>Association Based On :<br>Grid Hosting Capacity Document contextual model::Cadastre.Cadastre[0..*]<br>-----<br>Grid Hosting Capacity Document contextual model::UsagePointLocation.[]  |
| ##MXM## |        |                          |                                                                                                                                                                                                                             |

VoltageLevel

For a production unit or a generating unit, the high voltage level at the network side.

For other network equipment the voltage level.

!# shows all attributes of VoltageLevel.

Table !# - Attributes of Grid Hosting Capacity Document assembly model::VoltageLevel

| Order   | mult.  | Attribute name / Attribute type                | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|--------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0       | [1..1] | mRID<br><br>String                             | Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended.<br>For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. |
| 1       | [1..1] | highVoltageLimit<br><br>ESMP_Voltage           | The bus bar's high voltage limit                                                                                                                                                                                                                                                                                                                                                                               |
| 2       | [0..1] | name<br><br>String                             | The name is any free human readable and possibly non unique text naming the object.                                                                                                                                                                                                                                                                                                                            |
| 3       | [0..1] | identifier<br><br>UUID                         | A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.                                                                                                                                                                                                                                                                                                     |
| 4       | [0..1] | lowVoltageLimit<br><br>ESMP_Voltage            | The bus bar's low voltage limit                                                                                                                                                                                                                                                                                                                                                                                |
| 5       | [1..1] | baseVoltage.nominalVoltage<br><br>ESMP_Voltage | The power system resource's base voltage, expressed on a phase-to-phase (line-to-line) basis. Shall be a positive value and not zero.<br>--- The base voltage used for all equipment within the voltage level.                                                                                                                                                                                                 |
| ##MXM## |        |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                |

!# shows all association ends of VoltageLevel with other classes.

| Table !# - Association ends of Grid Hosting Capacity Document assembly model::VoltageLevel with other classes |        |                    |                                                                                                          |
|---------------------------------------------------------------------------------------------------------------|--------|--------------------|----------------------------------------------------------------------------------------------------------|
| Order                                                                                                         | mult.  | Class name / Role  | Description                                                                                              |
| 6                                                                                                             | [0..1] | Location           | Location of this power system resource.                                                                  |
|                                                                                                               |        |                    | Association Based On :                                                                                   |
|                                                                                                               |        | Location           | Grid Hosting Capacity Document contextual model::Location.Location[0..1]                                 |
|                                                                                                               |        |                    | -----<br>Grid Hosting Capacity Document contextual model::VoltageLevel.[]                                |
| 7                                                                                                             | [0..1] | UsagePointLocation | Location of this power system resource.                                                                  |
|                                                                                                               |        |                    | Association Based On :                                                                                   |
|                                                                                                               |        | UsagePointLocation | Grid Hosting Capacity Document contextual model::VoltageLevel.[]                                         |
|                                                                                                               |        |                    | -----<br>Grid Hosting Capacity Document contextual<br>model::UsagePointLocation.UsagePointLocation[0..1] |
| ##MXM##                                                                                                       |        |                    |                                                                                                          |

Datatypes

The list of datatypes used for the Grid Hosting Capacity Document assembly model is as follows:

- Status compound
- CEEDS\_ActivePower datatype
- ESMP\_DateTime datatype
- ESMP\_Length datatype
- ESMP\_Voltage datatype
- ESMPVersion\_String datatype
- ID\_String datatype
- MarketRoleKind\_String datatype, codelist RoleTypeList
- PartyID\_String datatype, codelist CodingSchemeTypeList
- UnitSymbol datatype, codelist UnitSymbol